Unbiased Estimators & Confidence Intervals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

Final Answer: -2.10. Working through this statistics step: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{44 - 47}{10/\sqrt{49}} \approx -2.10$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 3 standard deviations of the mean? Show your complete algebraic work.

Final Answer: 99.7%. Working through this statistics step:

By the empirical rule, about 99.7% of values lie within 3σ of μ.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 1 standard deviation of the mean? Show your complete algebraic work.

Final Answer: 68.3%. Working through this statistics step:

By the empirical rule, about 68.3% of values lie within 1σ of μ.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

(a) Sample mean: $\bar{x} = \frac{5090}{50} = 101.8$ g Unbiased sample variance:

$s^2 = \frac{1}{50-1}(518380 - 50(101.8)^2) = \frac{1}{49}(518380 - 518162) = \frac{218}{49} \approx 4.449$ g^2 (b) Standard error: SE $= \sqrt{\frac{s^2}{n}} = \sqrt{\frac{4.449}{50}} \approx 0.2983$.

$\bar{x} \pm 1.96(0.2983) = 101.8 \pm 0.585 = (101.2,\ 102.4)$ g (1 d. p.)

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A continuous dataset is grouped into class intervals. The interval $20 \le x < 35$ has a frequency of 45. Find the frequency density for this class.

Class width $w = 35 - 20 = 15$. Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{45}{15} = 3.0$. In a histogram, the height of the bar drawn on the interval $[20, 35)$ is 3.0.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

$142 \pm 2.576 \cdot \frac{20}{\sqrt{100}} = 142 \pm 5.15 = (136.85,\ 147.15)$. "99% confidence" means: if this process were repeated many times, 99% of the intervals produced would contain the true μ.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2-1, 4-2, 1-3 \rangle = \langle 1, 2, -2 \rangle$.
$\vec{AC} = \langle -1-1, 0-2, 5-3 \rangle = \langle -2, -2, 2 \rangle$. Step 2: Compute normal vector via cross product:
$\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k};\ 1, 2, -2;\ -2, -2, 2| = \mathbf{i}(4-4) - \mathbf{j}(2-4) + \mathbf{k}(-2-(-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2 \rangle$. Simplifying by scalar.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $ablaf = \lambda ablag$: $ablaf = \langle 2x, 4y \rangle$, $ablag = \langle 2x, 2y \rangle$. 1) $2x = 2\lambda x \Rightarrow 2x(1-\lambda) = 0$ 2) $4y = 2\lambda y \Rightarrow 2y(2-\lambda) = 0$ 3) $x^2 + y^2 = 1$ From (1), either $x = 0$ or $\lambda = 1$: • If $x = 0$: from constraint (3), $y^2 = 1 \Rightarrow y = \pm 1$. Points: $(0, 1)$ and $(0, -1)$. Value: $f(0, \pm 1) = 0 + 2(1)^2 = 2$. • If $\lambda = 1$: from (2), $2y(2-1) = 2y = 0 \Rightarrow y = 0$. From constraint (3).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5 \sqrt{9 - x^2} + \sqrt{9 - x^2} \right) dx$.

Split the integral using linearity: $\int_{-3}^{3} x^5 \sqrt{9 - x^2}\, dx + \int_{-3}^{3} \sqrt{9 - x^2}\, dx$. Part 1 · Symmetry Analysis: Let $g(x) = x^5 \sqrt{9 - x^2}$. Then $g(-x) = (-x)^5 \sqrt{9 - (-x)^2} = -x^5 \sqrt{9 - x^2} = -g(x)$. Because g is an odd function integrated over symmetric limits $[-3, 3]$: $\int_{-3}^{3} x^5 \sqrt{9 - x^2}\, dx = 0$. Part 2 · Geometric Area: The integrand $y = \sqrt{9 - x^2} \geq 0$ represents the upper half of the circle $x^2 + y^2 = 9$ with radius $r = 3$. The integral from.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the binomial expansion of $\frac{1}{\sqrt{4 - x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

Rewrite with negative fractional index and factor out 4: $(4 - x)^{-1/2} = 4^{-1/2} \left(1 - \frac{x}{4}\right)^{-1/2} = \frac{1}{2} \left(1 - \frac{x}{4}\right)^{-1/2}$
Apply $(1 + y)^n = 1 + ny + \frac{n(n-1)}{2} y^2 + \cdots$ with $n = -\frac{1}{2}$ and $y = -\frac{x}{4}$:
$\left(1 - \frac{x}{4}\right)^{-1/2} = 1 + \left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2} \left(-\frac{x}{4}\right)^2 + \cdots$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

Rate of change of salt: $\frac{dS}{dt} = \text{Rate In} - \text{Rate Out}$. Rate In $= (0.5 \text{ kg/L}) \times (4 \text{ L/min}) = 2 \text{ kg/min}$. Since volume remains constant at 100 L, salt concentration is $S(t)/100$.

Rate Out $= \left(\frac{S(t)}{100} \text{ kg/L}\right) \times (4 \text{ L/min}) = \frac{4S}{100} = 0.04S \text{ kg/min}$. Set up the linear ODE:

$\frac{dS}{dt} = 2 - 0.04S = -0.04(S - 50)$ Separate variables.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Express as a limit of a definite integral: $\int_0^\infty xe^{-x/2}\,dx = \lim_{t\to\infty}\int_0^t xe^{-x/2}\,dx$ Integrate by parts on $[0, t]$: Let $u = x \Rightarrow du = dx$, and $dv = e^{-x/2}dx \Rightarrow v = -2e^{-x/2}$. $\int_0^t xe^{-x/2}\,dx = [-2xe^{-x/2}]_0^t - \int_0^t(-2e^{-x/2})\,dx = -2te^{-t/2} + 0 + 2[-2e^{-x/2}]_0^t$ $= -2te^{-t/2} - 4(e^{-t/2} - 1) = 4 - e^{-t/2}(2t + 4)$ Take the limit as $t \to \infty$: By L'Hôpital's Rule.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty}\frac{(2x-1)^n}{n^2\cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty}\left|\frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}}\cdot\frac{n^2 4^n}{(2x-1)^n}\right| = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^2\cdot\frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x - 1 = 4$, giving $\sum_{n=1}^{\infty}\frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty}\frac{1}{n^2}$, which converges by the p-series test ($p = 2 > 1$). Endpoint $x = -3/2$: $2x - 1 = -4$, giving.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Let $u = 2x + 1$, so $du = 2\,dx \Rightarrow dx = \frac{1}{2}\,du$, and $x = \frac{u-1}{2}$. Change limits: $x = 0 \Rightarrow u = 1, \quad x = 4 \Rightarrow u = 9$ Transform the integral: $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx = \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}}\cdot\frac{1}{2}\,du = \frac{1}{4}\int_1^9 (u^{1/2} - u^{-1/2})\,du$ $= \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9 = \frac{1}{4}\left(\left(\frac{2}{3}(27) - 2(3)\right) - \left(\frac{2}{3}(1) - 2(1)\right)\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

(a) Under H_0, $X \sim \text{Po}(6)$. Calculate cumulative probabilities: $P(X = 0) = e^{-6} \approx 0.00248$ $P(X \leq 1) = e^{-6}(1 + 6) = 7e^{-6} \approx 0.01735$ $P(X \leq 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} \approx 0.06197$ Since $P(X \leq 1)0.05$, the critical region is $X \leq 1$ (i.e. $X = 0$ or 1). (b) Actual significance level = $P(X \leq 1 \mid \lambda = 6) = 7e^{-6} \approx 0.0174$ (or 1.74%). (c) Type II error occurs when we fail to reject H_0 when $\lambda = 3$. We fail to reject when.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

Factor 2 from the first two terms: $2\left(x^2 - \frac{5}{2}x\right) - 3 = 0$. Halve the coefficient of x: $\frac{1}{2}\left(-\frac{5}{2}\right) = -\frac{5}{4}$. Complete the square: $2\left(x - \frac{5}{4}\right)^2 - 2\left(\frac{25}{16}\right) - 3 = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8} - \frac{24}{8} = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 = \frac{49}{8}$. Divide by 2: $\left(x - \frac{5}{4}\right)^2 = \frac{49}{16}$. Take square roots.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$: $1 + 3 + 5 + \ldots + (2k - 1) = k^2$. 3. Inductive Step ($n = k + 1$): Show that $1 + \ldots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$: LHS $= [1 + 3 + \ldots + (2k - 1)] + (2k + 1)$. By the inductive hypothesis, substitute k^2.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 The Poisson Distribution

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x - 2)(x + 2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Further Integration Techniques

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

First find the components: $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$. The first derivative is $\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$. Differentiating with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2} + \frac{3}{2t^2}$. Then $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$. At $t = 2$: numerator is $\frac{3}{2} + \frac{3}{8} = \frac{15}{8}$; denominator is $2(2) = 4$. Thus.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Differential Equations

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

(a) Complete the squares for x and y: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 \Rightarrow (x - 3)^2 + (y + 2)^2 = 25$ The centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$. (b) The gradient of the radius from centre $(3, -2)$ to $P(6, 2)$ is: $m_{\text{rad}} = \frac{2 - (-2)}{6 - 3} = \frac{4}{3}$ The tangent is perpendicular to the radius, so its gradient is $m_{\text{tan}} = -\frac{3}{4}$. Using point-slope form with $P(6, 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s^2_{n-1} of the population variance σ^2?

a. $s^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ ✓

b. $s^2_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n}x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

Dividing by $n - 1$ removes the downward bias of the sample variance, ensuring that $E(s^2_{n-1}) = \sigma^2$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s_{n-1}^2.

a. 7.5 ✓

b. 6.0

c. 30

d. 9.0

$\bar{x} = 30/5 = 6$. $\sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 210 - 5(36) = 210 - 180 = 30$. Unbiased variance: $s_{n-1}^2 = \frac{30}{5-1} = \frac{30}{4} = 7.5$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?

a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$ ✓

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

$s_{n-1}^2 = \frac{n}{n-1} s_n^2 = \frac{16}{15}(16) \Rightarrow s_{n-1} = 4\sqrt{\frac{16}{15}} \approx 4.13$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$ ✓

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

Standard error is $\frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5$. Margin of error is $1.96(1.5) = 2.94$. Interval: $50 \pm 2.94 = [47.06, 52.94]$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9 ✓

b. 3

c. $\sqrt{3}$

d. 6

Since margin of error is proportional to $\frac{1}{\sqrt{n}}$, to reduce it by 3 requires $\sqrt{n_{\text{new}}} = 3\sqrt{n} \Rightarrow n_{\text{new}} = 9n$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the critical value z^* for a 99% confidence interval?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed interval, each tail contains 0.005 of the probability. The corresponding standard normal critical value is $z^* \approx 2.576$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. **Student's t-distribution with 9 degrees of freedom** ✓

b. Standard Normal Z-distribution

c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

Replacing σ with sample standard deviation s_{n-1} necessitates the t-distribution with degrees of freedom $df = n - 1 = 10 - 1 = 9$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How does the critical value t^* compare to z^* for the same confidence level and finite sample size?

a. $t^* > z^*$**, because the** t**-distribution has heavier tails to reflect estimation uncertainty** ✓

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

The t-distribution exhibits greater tail dispersion than the standard normal distribution to account for estimating σ from sample data, making $t^* > z^*$ for any finite degrees of freedom.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$ ✓

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

Standard error $= \frac{s}{\sqrt{n}} = \frac{2.0}{4} = 0.50$. Margin of error $= 2.131(0.50) = 1.0655$. Interval: $22.0 \pm 1.07 = [20.93, 23.07]$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which statement correctly conveys the meaning of a '95% confidence level'?

a. If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter ✓

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

The confidence level describes the long-run coverage rate of the interval estimation process over repeated random sampling.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval ✓

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

A hypothesized value outside a $(1 - \alpha)$ confidence interval is rejected at significance level α in a two-tailed test.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens ✓

b. The interval narrows

c. The interval width remains identical

d. The interval shifts to the right

Higher confidence requires capturing a larger portion of the sampling distribution, which increases the critical value and widens the margin of error.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Fourier series period 2π: $a_0 =$

a. $\frac{1}{\pi}\int_{-\pi}^{\pi} f(x),dx$

b. $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x),dx$ ✓

c. $\int_{-\pi}^{\pi} f(x)\cos x\,dx$

d. $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)\cos x\,dx$

Applying AA HL — Additional Topics principles, a■=(1/2π)∫f(x)dx. Therefore, the verified correct option is $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x),dx$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A transition matrix is $T = \left(\frac{4}{5}, \frac{3}{5}; \frac{1}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

a. $\frac{1}{10}$

b. $\frac{2}{5}$ ✓

c. 0

d. $\frac{1}{5}$

Working through this additional-topics step: Reading the matrix entry for $B \to B$ gives $\frac{2}{5}$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A transition matrix is $T = \left(\frac{4}{5}, \frac{2}{5}; \frac{1}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

a. $\frac{4}{5}$ ✓

b. $\frac{7}{10}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$

Working through this additional-topics step: Reading the matrix entry for $A \to A$ gives $\frac{4}{5}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

a. 1

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $\frac{3}{5}$ ✓

Working through this additional-topics step: Reading the matrix entry for $A \to B$ gives $\frac{3}{5}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A transition matrix is $T = \left(\frac{4}{5}, \frac{4}{5}; \frac{1}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

a. 1

b. $\frac{2}{5}$

c. $\frac{3}{5}$

d. $\frac{4}{5}$ ✓

Working through this additional-topics step: Reading the matrix entry for $B \to A$ gives $\frac{4}{5}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 The Poisson Distribution

Describe the locus $|z-(4-2i)|=4$ on the Argand diagram.

a. Circle, centre $(4,-2)$, radius 16

b. Circle, centre $(-4,2)$, radius 4

c. Line through $(4,-2)$

d. Circle, centre $(4,-2)$, radius 4 ✓

Working through this additional-topics step: $|z-z_0|=r$ is a circle centred at $z_0=(4,-2)$ with radius 4.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Further Integration Techniques

A 2-state Markov chain has transition matrix $T=\left(\frac{9}{10},\frac{1}{10};\frac{3}{10},\frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$ ✓

Working through this additional-topics step: Steady state: $\pi_1=\frac{3}{10}/(\frac{1}{10}+\frac{3}{10})=\frac{3}{4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Differential Equations

A system starts in state A. The transition matrix (rows/cols A, B) is $T=\left(\frac{4}{5},\frac{1}{5};\frac{3}{5},\frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

a. $\frac{3}{5}$

b. $\frac{7}{10}$

c. $\frac{19}{25}$ ✓

d. $\frac{2}{5}$

Working through this additional-topics step: $\pi_1=\pi_0T=\frac{4}{5},\frac{1}{5};\ \pi_2=\pi_1T\Rightarrow P(A)=\frac{19}{25}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find the value of k for which $(4, 3; 4, k)$ is singular (has no inverse).

a. 1

b. -3

c. **3** ✓

d. 5

Working through this additional-topics step: $\text{Singular} \Leftrightarrow \det = 0 : 4k - 3(4) = 0 \Rightarrow k = \frac{12}{4} = 3$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{3}{10}, \frac{1}{2}$. Find $\text{Var}(X)$.

a. $\frac{61}{100}$ ✓

b. $\frac{121}{100}$

c. $\frac{73}{100}$

d. $\frac{4}{5}$

Working through this statistics step: $E(X) = \frac{13}{10},\ E(X^2) = \frac{23}{10} \Rightarrow \text{Var}(X) = \frac{61}{100}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Matrices (AI HL)

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

a. 0.2231

b. 0.1255

c. 0.2510

d. **0.3347** ✓

$\lambda = np = 150 \times \frac{1}{100} = 1.5.\ P(X = 1) \approx \frac{e^{-1.5}1.5^1}{1!} \approx 0.3347$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Given $\sin A = \frac{8}{17}$, $\cos A = \frac{15}{17}$, $\sin B = \frac{3}{5}$, $\cos B = \frac{4}{5}$ (both acute), find $\sin(A + B)$.

a. $\frac{69}{85}$

b. $\frac{82}{85}$

c. $\frac{77}{85}$ ✓

d. $\frac{16}{17}$

Working through this additional-topics step: $\sin(A + B) = \sin A \cos B + \cos A \sin B = \frac{77}{85}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.03 Markov Chains (AI HL)

In a survey of 150 people, 74 are category A and 74 have trait X, with 12 being both A and X. Find $P(A|X)$.

a. $\frac{3}{5}$

b. $\frac{137}{150}$

c. $\frac{6}{37}$ ✓

d. $\frac{9}{37}$

Working through this statistics step: $P(A|X) = \frac{n(A \cap X)}{n(X)} = \frac{12}{74} = \frac{6}{37}$

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